



Fig. 4: Author's prototype



Fig. 5: Readings before pressing S2



Fig. 6: Readings after pressing S2 (while it is kept pressed)



Fig. 7: Readings showing values of d and D

To measure the distance from point A to C, press S2 and keep it pressed till you reach point B. Release S2 to stop the measurement; d/D values (instantaneous/overall distance) on LCD will be shown as 10/10. The same values will be stored on the microSD card. Now, press S2 again till you reach point C. Release S2 and note the d/D values. You will see 20/30 on the LCD.

Finally, when all measurement readings are taken, press S3 to close the file and halt the system. Reset Arduino to restart the system. Author's prototype is shown in Fig. 4.

The log file gets stored on the microSD card. Every time you start the device, it will create a new log file by the names data0.csv, data01.csv, data02.csv and so on as given below:

Sl Date Time Latitude Longitude
Dist. Dist0

```
0 20/3/16 23:1:18 24.088750
82.648200 0.0000 0.0000
1 20/3/16 23:1:21 24.088750
82.648200 0.0000 0.0000
2 20/3/16 23:1:24 24.088750
82.648200 0.0000 0.0000
3 20/3/16 23:1:29 24.088750
82.648200 0.0000 0.0000
```

EFY note. 1. The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

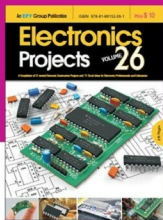
2. The project was tested at EFY Lab on Arduino Uno with an external 16MHz crystal and was found to be working well. **EFY**



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